Backtest Overfitting Statistics under Serial Dependence

A Computational Replication and Empirical Stress Test: research summary

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Question. PSR, the 2014 DSR and CSCV/PBO are often read as hurdles ("DSR > 0.95", "PBO ≤ 0.05"). How do they behave as decision rules when returns are serially dependent, heteroskedastic, heavy tailed or cross-correlated, and when a historical search winner is tested against a real benchmark?

What was done. Independent, test-driven reimplementation of PSR, MinTRL, DSR, MinBTL and CSCV/PBO with page-level provenance; selected numerical examples from the four source papers reproduced. Zero-edge Monte Carlo (2000 datasets per cell) under IID, AR(1), GARCH(1,1), Student-t and equicorrelated returns. Circular block bootstrap of daily Kenneth French 49-industry returns, 1971–2026, centred to zero edge, including a sub-period split at 1999. A search over 93 moving-average rules on the Fama–French market factor (1972–2026, 5 bps per position change), tested against buy-and-hold with a paired HAC Sharpe-difference test, Bonferroni and Hansen's SPA, plus exploratory lag, cost, sub-period and walk-forward checks. Every saved output was regenerated from the current code and matched the archived results.

Main findings.

1. **PSR and dependence (confirms known theory).** At zero Sharpe, uncorrected PSR(0)> 0.95 rejects 11.85% [10.47, 13.35] under AR(1) $\phi = 0.3$; the large-sample prediction $1-\Phi(1.645/\sqrt{F})$ with $F = (1+\phi)/(1-\phi)$ gives 11.37%. GARCH, Student-t and correlated designs stay at 5% (as expected at $\mathrm{SR}^* = 0$). A 7-lag Newey–West variance reduces the AR(1) rate to 7.10%, not 5%.
2. **PSR on real data (new evidence, conditional).** With each period centred, resampled 1971–1998 industry returns give 8.35% [7.17, 9.65] and 1999–2026 give 3.85% [3.05, 4.79]: the error runs in both directions and tracks lag-1 autocorrelation ($+0.157$ vs -0.026 on average). A long-run-variance approximation lands within 0.1 points in both periods. The full-window rate (about 6%) is a mixture.
3. **The 2014 DSR is not a 5% test.** On the best of 100 IID zero-skill strategies it rejects 0.05% [0.00, 0.28] (PSR: 99.15%). Acceptance rises with cross-strategy correlation (2.60% at $\rho = 0.7$; 5.05% with the Eq. 9 implied $N = 30.7$), because V omits the common component.
4. **PBO under exchangeable nulls.** Mean PBO stays near 1/2 (or 25/49) in every zero-edge design; such nulls do not test PBO's power, which was not measured.
5. **Case study.** Selected rule 15/225: Sharpe 0.588 vs market 0.458, DSR 0.998, PBO 0.208. Paired HAC difference $+0.130$ (s.e. 0.099, one-sided $p = 0.094$); 0 of 93 rules pass Bonferroni; SPA $p = 0.504$. By a Normal approximation holding the s.e. fixed, 80% power needs a difference of 0.246 (0.406 with Bonferroni). Walk-forward re-selection (exploratory) earns 0.466 vs the market's 0.536 over 1982–2026 (corrected for switch costs).

Decision	Statistic	Does not establish
One pre-specified Sharpe above a hurdle	PSR	Robustness to serial dependence
Search winner better than best of N noise trials	DSR (2014)	A 5% error rate; skill vs a benchmark
Rankings persist out of sample	PBO	A probability of future loss
Move capital from benchmark to strategy	none of these	needs a search-aware benchmark test with power

Contribution and novelty limits. Replication with traceable provenance; quantified operating characteristics of the 2014 DSR and of PSR on real daily data; an explicit contrast between search-adjusted zero-skill inference and benchmark-relative inference, with power. Not claimed: a new statistic, first identification of the serial-dependence problem (Lo 2002; Ledoit and Wolf 2008), or any profitable strategy. The 2026 DSR framework of López de Prado and Porcu, examined only through its abstract, may subsume the DSR findings.

Status of evidence. Designs and predictions were recorded before runs; from the case study on they were committed to version control before running. The evidence rule, the sub-period prediction and all robustness checks were fixed after earlier results were known and are labelled exploratory or post hoc. One earlier sub-period result was withdrawn after an audit found the periods were not centred.

Main limitations. One data vendor and market; $\mathrm{SR}^* = 0$ only for PSR; fixed bootstrap block lengths; idealized same-close execution on a research portfolio; HAC assumptions unverified for 54 years of data; external data access/redistribution terms; both input archives now have timestamped, hash-verified recovery URLs.